

TASK 05-50-27-200-801-A
EFFECTIVITY: ALL

1. Landing Gear Down Overspeed - Inspection/Check

A. General

- (1) This task gives the procedure that you must do when at least one of the events below occurs:
 - Retraction of the landing gear at an airspeed higher than the airspeed limit for landing gear operation.
 - Extension of the landing gear at an airspeed higher than the airspeed limit for landing gear operation.
 - Flight with the landing gear down and locked at an airspeed higher than the airspeed limit for flight with landing gear down.
- (2) This maintenance technician is necessary to do this task:
 - 1 - Mechanical Systems
- (3) This maintenance technician is necessary to do this task:
 - 1 - Electrical/Avionics Systems

B. References

REFERENCE	DESIGNATION
AMM TASK 32-00-00-840-801-A/200	Landing Gear - Prepare for Maintenance
AMM TASK 32-00-00-840-802-A/200	Landing Gear - Restore after Maintenance
AMM TASK 32-12-00-820-801-A/200	Main-Landing-Gear Doors - Adjustment
AMM TASK 32-12-01-860-801-A/200	Main-Landing-Gear Inboard Door (open) - Configuration
AMM TASK 32-12-01-860-802-A/200	Main-Landing-Gear Inboard Door (close) - Configuration
AMM TASK 32-22-01-820-801-A/200	Nose-Landing-Gear Forward-Door Rods Linkage - Adjustment
AMM TASK 32-22-02-860-801-A/200	Nose-Landing-Gear Forward Door (open) - Configuration
AMM TASK 32-22-02-860-802-A/200	Nose-Landing-Gear Forward Door (close) - Configuration
AMM TASK 32-22-03-820-801-A/200	Nose-Landing-Gear Aft-Door Rods Linkage - Adjustment
AMM TASK 32-30-01-710-801-A/500	Landing-Gear Extension-and-Retracton System (On Jacks) - Operational Test
AMM TASK 32-35-00-710-801-A/500	Emergency-Extension System - Free Fall (On Jacks) - Operational Test

C. Overspeed Landing Gear Operation - Event Assessment

SUBTASK 200-002-A

- (1) The IAS limits for flight with landing gear in transit or locked down are given in section 2 of the AFM and named as follow:

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- V_{LO} (e) is the maximum speed that the landing gear can be safely extended.
 - V_{LO} (r) is the maximum speed that the landing gear can be safely retracted.
 - V_{LE} is the maximum speed that the airplane can be safely flown with the landing gear down and locked.
- (2) The maximum value speed is as follows to give the operational envelope for flight with the landing gear in transit or locked down:
- (a) The V_{LO} (e) value is 250 KIAS.
 - (b) The V_{LO} (r) value is 220 KIAS.
 - (c) The V_{LE} value is 265 KIAS.
- (3) The purpose of this section is to provide the operator with all information to analyze the flight data, determine the required inspections, and restore the aircraft airworthiness.
- (4) For this event calculation, the parameters below must be selected from the flight data:
- Hours (GMT);
 - Minutes (GMT);
 - Seconds (GMT);
 - Indicated air speed;
 - Main landing gear position (Left or Right);
 - Nose landing gear position.

D. Overspeed Landing Gear Operation - Exceedance Detection

SUBTASK 200-003-A

- (1) Identify the flight period when the event occurred in the flight data.
- (2) Examine the value of airspeed and landing gear control-lever position in the interval when the event occurred.
- (3) Identify the samples where the airspeed (IAS) was higher than:
 - (a) The V_{LO} (e) value is 250 KIAS.
 - (b) The V_{LO} (r) value is 220 KIAS.
 - (c) The V_{LE} value is 265 KIAS.
- (4) If speed value is higher than values above, do the [SUBTASK 220-002-A](#).

E. Job Set-Up

SUBTASK 840-002-A

WARNING: MAKE SURE THAT THE AIRCRAFT IS IN A SAFE CONDITION BEFORE YOU DO THE MAINTENANCE PROCEDURES. THIS IS TO PREVENT INJURY TO PERSONS AND/OR DAMAGE TO THE EQUIPMENT.

- (1) Do the procedure to make the aircraft safe for maintenance of the landing gear (AMM TASK 32-00-00-840-801-A/200).

SUBTASK 010-002-A

- (2) Open the nose-landing-gear doors (AMM TASK 32-22-02-860-801-A/200).

SUBTASK 010-003-A

- (3) Open the main-landing-gear inboard doors (AMM TASK 32-12-01-860-801-A/200).

F. Overspeed Landing Gear Operation - Inspection

SUBTASK 220-002-A

- (1) Do these tasks to adjust the gaps and steps:
 - AMM TASK 32-12-00-820-801-A/200;
 - AMM TASK 32-22-01-820-801-A/200;
 - AMM TASK 32-22-03-820-801-A/200.
- (2) Do these operational tests:
 - AMM TASK 32-30-01-710-801-A/500;
 - AMM TASK 32-35-00-710-801-A/500.
- (3) In case there are findings, send the flight data and the inspection results to EMBRAER and work together with EMBRAER for the necessary actions.

G. Job Close-Up

SUBTASK 940-002-A

- (1) Remove from the work area all tools, equipment, and materials that you used.

SUBTASK 410-002-A

- (2) Close the nose-landing-gear doors (AMM TASK 32-22-02-860-802-A/200).

SUBTASK 410-003-A

- (3) Close the main-landing-gear inboard doors (AMM TASK 32-12-01-860-802-A/200).

SUBTASK 840-003-A

- (4) Do the procedure to restore the aircraft after the maintenance of the landing gear (AMM TASK 32-00-00-840-802-A/200).

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